

Activity 4: ER Relationships

CSCI 340 / DATABASE DESIGN

Choose the concepts, explain the constraints, draw the model

Name and collaborators

Work individually or in a group. Explain your choices from the scenarios and state any additional assumptions. Leave time to compare answers in Part E, even if you work alone.

Use the lecture's arrow notation for diagrams. Use entity names in your rectangles; you do not need to choose keys or write SQL. Blank spaces are for explanations and sketches.

Part A: What are we modeling?

A college records students and their names and email addresses, instructors and their names and email addresses, and scheduled course sections with titles and meeting rooms. Students enroll in sections, and instructors teach sections. Several sections can share a course title. Model only the current term.

A1. Identify the entity sets and their attributes

Identify three entity sets and give one example entity in each. Place the descriptive attributes with the appropriate entity sets. Explain why a course title alone would not distinguish individual sections.

A2. Identify the relationships

Name the two relationships and identify the entity sets each connects. Choose one relationship and explain why it represents an association in this model. Then explain why an email address is an attribute here rather than a separate entity set.

Part B: How many partners are allowed?

For B1–B3, name the relationship and state its cardinality. Justify your answer.

B1. Personal lockers at a gym

Member is a gym member; **Locker** is a particular physical locker. Members may rent a personal locker for ongoing use. A member can hold one rental at a time, and that locker is reserved for that member alone. Unrented lockers remain in inventory. Model current rentals.

B2. Home departments at a company

Employee is a current employee; **Department** is an organizational unit. Model each employee's single home department on the company roster. A department brings together employees who do different jobs; a newly formed department may still be awaiting its first employee.

B3. Enrollment at a college

Student is a registered student; **Section** is one scheduled offering of a course in the current term. Students choose sections to build their schedules, and each section has a class roster. The system includes students who have not enrolled yet and sections awaiting their first students. Model enrollment.

For B2, also describe the cardinality in reverse. Does reversing your reading change the relationship itself?

Part C: Build and explain an ER diagram

Use **Student**, **Instructor**, and **Section** from Part A, and the enrollment rules from B3. Each section is taught by exactly one instructor. An instructor may teach several sections or none.

C1. Draw one combined model

Connect the three entity sets using the two relationships from A2. Include one descriptive attribute for each entity set. Use arrows for maximum cardinality and single or double lines for participation. Explain the placement of one arrow and one double line.

C2. Add and justify a relationship attribute

The college records `final_grade` for each student in each section they complete. Add this attribute to your diagram with a dashed connection. Explain why a single `final_grade` on **Student** or a single `final_grade` on **Section** would lose information. Give one concrete example for each.

Part D: Who must participate?

For D1–D3, distinguish minimum participation from maximum cardinality. Explain each answer using a scenario rule, not just the current records.

D1. Revisit the gym and company

For each side of B1 and B2—**Member**, **Locker**, **Employee**, and **Department**—state whether participation is total or partial. Identify the phrase that supports each choice.

D2. Change a college rule

Suppose every registered student must now enroll in at least one section. Empty sections are still allowed. Which participation line in C1 changes? Does this change the maximum cardinality of enrollment? Explain.

D3. Evaluate a classmate's inference

Today, every instructor happens to teach exactly one section. A classmate concludes that teaches is one-to-one and participation is total on both sides. Evaluate both claims against the original Part C rules. Which part of the participation claim is actually supported?

Part E: Compare and reflect

Compare your work with another individual or group. Identify whom you compared with and one difference or shared choice about an entity, relationship, arrow, participation line, or attribute. Explain why you revised or retained your answer, citing the scenario. If you disagree, identify the assumption to clarify.